

1. TYPES OF DATA

Data can be categorized based on its nature and characteristics

TYPES

- * Qualitative Data - Descriptive non-numeric (e.g., names, genders)
- * Discrete Data - Countable, whole numbers
- * Continuous Data - Measurable on a continuum (e.g., height)
- * Categorical data - Grouped into categories
- * Ordinal Data - Ordered categories (e.g., ranking)

3. FORMATS OF DATA

Data can be stored and represented in different format

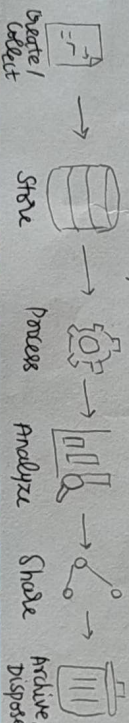
FORMATS

- * Structured Data (e.g., tables)
- * Semi-structured data (e.g., JSON, XML)
- * Unstructured data (e.g., text, images, videos)

6. DATA LIFECYCLE (A KEY PERSPECTIVE)

Data goes through a lifecycle from creation to disposal

STAGES



8. CHALLENGES IN MANAGING DATA

Organizations must address common data challenges

CHALLENGES

- Data overload
- Data Privacy & Security
- Data Silos
- Data Quality Issues
- Data Integration & Interoperability
- Cost & Storage & Processing
- Skill Gap

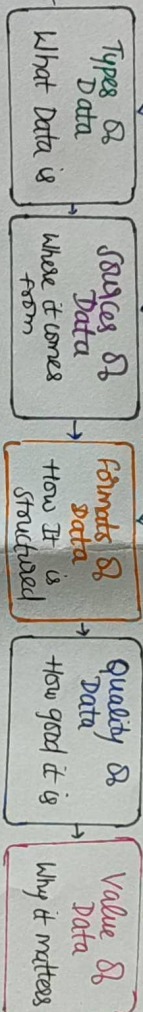
SUMMARY:

The facets of data - its types, sources, format, quality & value - work together across the data lifecycle to drive insights, support decisions and create impact. Managing these facets effectively leads to trustworthy, useful and responsible data.

UNDERSTANDING THE DIFFERENT DIMENSIONS, TYPES AND ROLES OF DATA

Exploring the nature, variety and significance of data in decision making, analysis and real world application

THESE FACETS WORK TOGETHER



5. VALUE OF DATA

High-quality data delivers meaningful value

- Types of Data: What Data is
- Sources of Data: Where it comes from
- Formats of Data: How it is structured
- Quality of Data: How good it is
- Value of Data: Why it matters

7. ROLE OF DATA

Data plays a central role across industries & functions

ROLES

- Identifies Risks and Opportunities
- Supports Strategic Planning
- Enhances Customer Experience
- Improves Efficiency
- Drives Innovation
- Provides Better Decisions

9. BEST PRACTICES

Follow these practices to leverage data efficiency

PRACTICES

- Define clear objectives
- Ensure data governance
- Maintain data quality
- Use the right tools
- Foster a data-driven culture
- Enable automation
- Supports Reporting & Visualization
- Enables compliance & Governance

2. SOURCES OF DATA

Data is collected from various origins and channels

SOURCES

- * Internal sources - within an organization (e.g., databases, logs)
- * External sources - outside the organization (e.g., surveys, APIs, ...)
- * Primary data - collected first hand for a specific purpose
- * Secondary data - already collected by someone else

4. QUALITY OF DATA

The usefulness of data depends on its quality

DIMENSIONS

- * Accuracy - correct and error-free
- * Completeness - No important data is missing
- * Consistency - uniform across system and time
- * Timeliness - available when needed
- * Validity